

Simultaneous B_1^+ and T_1 estimation in multi-pathway multi-echo (MPME) imaging: forward model and fundamental limits

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Abstract

We derive the steady-state signal model of the multi-pathway multi-echo (MPME) sequence of Cheng et al. [1] and analyse the joint estimation of the transmit field B_1^+ and the longitudinal relaxation time T_1 through the Fisher information and the Cramér–Rao lower bound (CRLB). The central difficulty is a property of the *forward model*: for small flip angles and $T_R \ll T_1$ every pathway amplitude is invariant, to leading order, under $(M_0, B_1^+, T_1) \mapsto (M_0/\kappa, \kappa B_1^+, T_1/\kappa^2)$. Protocols whose flip angles are all small therefore have a severely ill-conditioned information matrix: the parameters remain identifiable in principle, but separating them is extremely noise-sensitive (log- T_1 bound of order one at realistic SNR). The near-360° second pulse of the published protocol lifts the invariance through the 2π wrap of the rotation, reducing the condition number from 5×10^5 to 173. What remains is a finite bound ($40.5 \sigma/M_0$ on the standard deviation of log T_1) set by the data. The published *analytic inverse* adds an estimator-specific loss: in Monte Carlo its T_1 standard deviation is 3.2–7.1 times the bound, with large bias and undefined estimates at moderate SNR. A joint maximum-likelihood fit of the same model, with the physical constraint $R_2' \geq 0$ enforced, attains the bound (ratio 0.99–1.02) at the SNRs where its assumptions hold, and comes close at lower SNR where the constraint is often active. For magnitude data we show that the published protocol has an exact alias inside the physiological B_1^+ range, because magnitudes lose the sign of $\sin \alpha$; the relative phase of the two FID images supplies the missing bit, and with it a magnitude least-squares fit performs like complex maximum likelihood, whereas the analytic inverse, using the same data, remains well above the bound.

How to read this note

Section 1 builds the forward model from the configuration-state (extended phase graph) description up to the measured voxel signal. Section 2 sets up the estimation-theoretic tools. Section 3 proves that each scan depends on the flip angle and T_1 only through two invariants (Theorem 1), derives the small-angle scaling invariance that underlies the B_1^+/T_1 problem, explains how the published protocol escapes it, and identifies the exact global aliases. Section 4 gives the bounds and the conditioning of the information matrix. Section 5 answers the question *is the limitation in the forward model or in the inverse?* by comparing the analytic reconstruction and a maximum-likelihood fit against the bound. Section 6 treats magnitude data and explains why a magnitude fit needs the FID-sign branch test while complex maximum likelihood does not. Section 7 derives exact Jacobians by implicit differentiation of the steady state, which make both fits 10–34 times faster. Table 7 summarises the answer.

All numbers are reproducible with the code in the accompanying repository (Appendix A).

1 Forward model

1.1 Sequence

The MPME sequence is a 3D gradient-echo sequence without RF spoiling, with a constant RF phase φ and repetition time T_R . Only the readout gradient G_x is unbalanced; its net area per repetition, $A = \int_0^{T_R} G_x dt$, dephases the magnetisation by many cycles across a voxel (the *fully dephased* regime). Two acquisitions $i = 1, 2$ use the same gradient waveform with nominal flip angles $\alpha_1^{\text{nom}} = 15^\circ$ and $\alpha_2^{\text{nom}} = 330^\circ$ and $T_R = 25$ ms [1]. The actual flip angles are

$$\alpha_i = B_1^+ \alpha_i^{\text{nom}}, \quad (1)$$

so B_1^+ scales all pulses by the same factor.

Each of three readout windows of alternating polarity sweeps the cumulative gradient moment through three adjacent integer multiples of A (lobe areas $+1.5 : -3 : +3 : -3 : +0.5$ in units of $-A$), sampling three *pathways* (a two-pathway predecessor is described in [2]) $k \in \{+1, 0, -1\}$ spaced by $1/\text{BW} \approx 2$ ms. A state of order k created by an RF pulse is refocused when the moment accumulated since the pulse equals $-kA$. This gives each pathway three echo times $t_{ijk} \in (0, T_R)$ (in ms, assuming 4.5 ms from the pulse to the first echo): $k = +1$: 4.5, 14.5, 16.5; $k = 0$: 6.5, 12.5, 18.5; $k = -1$: 8.5, 10.5, 20.5.

1.2 Configuration states

Let $M^+ = M_x + iM_y$ and let $\psi \in [0, 2\pi)$ be the intravoxel phase imparted by one repetition of the unbalanced gradient. Expand

$$M^+(\psi) = \sum_k F_k e^{ik\psi}, \quad M_z(\psi) = \sum_k Z_k e^{ik\psi}, \quad Z_{-k} = \overline{Z_k}. \quad (2)$$

One repetition of the operators acts as follows [4, 3].

RF pulse (flip α , phase φ ; rotation $R = R_z(\varphi)R_x(\alpha)R_z(-\varphi)$):

$$F'_k = \cos^2 \frac{\alpha}{2} F_k + e^{2i\varphi} \sin^2 \frac{\alpha}{2} \overline{F_{-k}} - ie^{i\varphi} \sin \alpha Z_k, \quad (3)$$

$$Z'_k = -\frac{i}{2} e^{-i\varphi} \sin \alpha F_k + \frac{i}{2} e^{i\varphi} \sin \alpha \overline{F_{-k}} + \cos \alpha Z_k. \quad (4)$$

Relaxation over T_R , with $E_1 = e^{-T_R/T_1}$ and $E_2 = e^{-T_R/T_2}$:

$$F_k \rightarrow E_2 F_k, \quad Z_k \rightarrow E_1 Z_k \ (k \neq 0), \quad Z_0 \rightarrow E_1 Z_0 + M_0(1 - E_1). \quad (5)$$

Gradient $F_k \rightarrow F_{k+1}$. Free precession by $\Delta\omega$ for a time t multiplies M^+ by $e^{i\Delta\omega t}$ (same sense as the gradient twist).

1.3 Steady state

Write $a_k(\alpha, T_1, T_2, T_R)$ for the steady-state amplitude of F_k immediately after the RF pulse, per unit M_0 . It can be computed exactly without truncating the state space: an isochromat with total precession $\beta = \psi + \Delta\omega T_R$ per repetition has the post-pulse steady state

$$(I - R R_z(\beta) \text{diag}(E_2, E_2, E_1)) \mathbf{M}^+ = (1 - E_1) R \hat{\mathbf{z}}, \quad (6)$$

and F_k is the k -th Fourier coefficient of $M^+(\psi)$ over ψ .

Proposition 1 (Off-resonance only adds phase). *In the fully dephased regime,*

$$a_k(\Delta\omega) = a_k(0) e^{ik\Delta\omega T_R};$$

in particular $|a_k|$ does not depend on $\Delta\omega$.

Proof. By (6) the steady state depends on ψ and $\Delta\omega$ only through $\beta = \psi + \Delta\omega T_R$, so $M_{\Delta\omega}^+(\psi) = M_0^+(\psi + \Delta\omega T_R)$. Shifting a periodic function by θ multiplies its k -th Fourier coefficient by $e^{ik\theta}$. $\square$

For the two lowest pathways the steady-state magnitudes have closed forms (unbalanced FISP and PSIF), which we verified against (6) to 10^{-6} :

$$\begin{aligned} |a_0| &= \left| \tan \frac{\alpha}{2} \left[1 - \frac{(E_1 - \cos \alpha)(1 - E_2^2)}{r} \right] \right|, \\ |a_{-1}| &= \left| \frac{\tan \frac{\alpha}{2}}{E_2} \left[1 - \frac{(1 - E_1 \cos \alpha)(1 - E_2^2)}{r} \right] \right|, \\ r^2 &= p^2 - q^2, \quad p = 1 - E_1 \cos \alpha - E_2^2(E_1 - \cos \alpha), \quad q = E_2(1 - E_1)(1 + \cos \alpha). \end{aligned} \quad (7)$$

The PSIF expression in the literature is the echo amplitude at $t = T_R$, hence the factor $1/E_2$ for the post-pulse state.

1.4 Echo evolution and the voxel signal

Pathway k read at time t after the pulse carries, for one isochromat, the off-resonance phase $\Delta\omega(t + kT_R)$ (Proposition 1 plus precession since the pulse). Averaging over a Lorentzian intravoxel frequency distribution of half-width R'_2 , $\langle e^{i\delta\tau} \rangle = e^{-R'_2|\tau|}$, gives the signal of scan i , echo j , pathway k :

$$S_{ijk} = M_0 a_k(B_1^+ \alpha_i^{\text{nom}}, T_1, T_2, T_R) e^{-t_{ijk}/T_2} e^{-R'_2|t_{ijk} + kT_R|} e^{i(\phi_0 + \Delta\omega(t_{ijk} + kT_R))} \quad (8)$$

with $T_2^* = 1/(1/T_2 + R'_2)$. For $k \geq 0$ the magnitude decays at $T_2^{-1} + R'_2$; for $k < 0$ and $t < |k|T_R$ at $T_2^{-1} - R'_2$ (a spin echo forming at $t = |k|T_R$).

Remark 1 (Relation to the published equations). Eq. 1 of [1], $S = F^+ e^{-(R_2 \pm R'_2)TE}$ (+ for $k \geq 0$), has the same echo-time dependence as (8); its intercept is $F_k^+ = a_k e^{-R'_2|k|T_R}$. The extra factor depends only on $|k|$, so every relation between the $\pm k$ states used in [1] (their Eqs. 4–13), and the mixing factor, are unchanged. We verified numerically that their Eqs. 7, 16 and 17 hold exactly on the steady states of (6) (Eq. 16 read as $T_1 = T_R / \ln[(Xc - 1)/(X - c)]$).

1.5 Imaging and noise

Image formation $y_{ijk} = \mathcal{M}_i \mathcal{F}(C_c S_{ijk}) + n_{ijk}$ (coil maps C_c , Fourier transform $\mathcal{F}$, sampling $\mathcal{M}_i$; scan 2 covers only the central $25\% \times 25\%$ of k_y – k_z) is linear. For the voxel-wise analysis below we model the coil-combined complex images as $S_{ijk} + n_{ijk}$ with independent circular Gaussian noise of standard deviation σ per real and imaginary part.

2 Estimation-theoretic framework

Collect the unknowns of one voxel as

$$\theta = (\log M_0, \log B_1^+, \log T_1, \log T_2, \log T_2^*, \Delta\omega, \phi_0). \quad (9)$$

For the complex Gaussian model the Fisher information is

$$\mathcal{I}(\theta) = \frac{1}{\sigma^2} \sum_{i,j,k} \text{Re} \left[\overline{\partial_\theta S_{ijk}} \partial_\theta S_{ijk}^\top \right], \quad \text{Cov}(\hat{\theta}) \succeq \mathcal{I}^{-1} \quad (10)$$

for any unbiased estimator (more precisely, any estimator that is locally unbiased at θ). Biased estimators, including constrained, regularised and learned ones, are not covered: they can have smaller variance and smaller mean-squared error, at the price of a bias that the CRLB does not quantify (for those, a Bayesian bound such as the van Trees inequality is the appropriate tool). Because positive parameters enter through their logarithm, $\sqrt{(\mathcal{I}^{-1})_{ll}}$ bounds the standard deviation of $\log \hat{\theta}_l$, which equals the relative standard deviation of $\hat{\theta}_l$ only to first order (Section 2.1). With σ expressed in units of M_0 every bound scales as $\sigma/M_0 = 1/\text{SNR}(M_0)$; we report bounds *per unit* σ/M_0 . The brightest MPME image has $|S| \approx 0.088 M_0$, so the image SNR is about $0.088 \text{SNR}(M_0)$.

We use three diagnostics of conditioning:

- the *normalised information* $\tilde{\mathcal{I}} = D^{-1/2} \mathcal{I} D^{-1/2}$, $D = \text{diag } \mathcal{I}$, its condition number $\kappa(\tilde{\mathcal{I}})$, and its smallest eigenvalue λ with eigenvector u . The eigenvector lives in normalised coordinates; the corresponding direction in the original coordinates is $w \propto D^{-1/2} u$, the solution of the generalised eigenproblem $\mathcal{I} w = \lambda D w$. Conversely, a near-null direction v of $\mathcal{I}$ appears in normalised coordinates as $u \propto D^{1/2} v$;
- the *variance inflation* $(\mathcal{I}^{-1})_{ll} \mathcal{I}_{ll} \geq 1$, the factor by which the bound on θ_l grows because the other parameters are unknown;
- the *fixed-parameter bias*: if θ_m is held at a wrong value $\theta_m + \delta$ and the rest is fitted, the estimates move by $\delta \theta_r = -\mathcal{I}_{rr}^{-1} \mathcal{I}_{rm} \delta$ to first order.

Jacobians are obtained by automatic differentiation through (6)–(8) and checked against finite differences.

2.1 Why logarithmic parameters

The five quantities $M_0, B_1^+, T_1, T_2, T_2^*$ are positive *scale* parameters: they carry units (or, for B_1^+ , multiply a quantity that does), are only meaningful relative to other scales in the problem, and are naturally compared by ratios. Their logarithms are the natural coordinates for five reasons, of different weight.

1. The model depends on them only through dimensionless ratios. Rewrite (8) as

$$\frac{S_{ijk}}{M_0} = G \left(B_1^+ \alpha_i^{\text{nom}}, \frac{T_R}{T_1}, \frac{T_R}{T_2}, \frac{t_{ijk}}{T_2}, R_2' t_{ijk}, \Delta \omega t_{ijk} \right), \quad (11)$$

since $E_1 = e^{-T_R/T_1}$, $E_2 = e^{-T_R/T_2}$ and the echo terms depend only on products of rates and times. A change of units, or a common rescaling of all times, is an additive shift in log coordinates and leaves G unchanged. The derivatives with respect to the log parameters are dimensionless *elasticities*, comparable across parameters:

$$\begin{aligned} \frac{\partial S}{\partial \log M_0} &= S, & \frac{\partial S}{\partial \log B_1^+} &= \alpha_i \frac{\partial S}{\partial \alpha_i}, \\ \frac{\partial S}{\partial \log T_1} &= -\frac{T_R}{T_1} \frac{\partial S}{\partial (T_R/T_1)}, & \frac{\partial S}{\partial \log T_2} &= \frac{t}{T_2} S - \frac{T_R}{T_2} \frac{\partial S}{\partial (T_R/T_2)} \Big|_{\text{steady state}}. \end{aligned} \quad (12)$$

(The T_2 elasticity is written at fixed R'_2 ; with T_2^* held fixed instead, as in θ , $R'_2 = 1/T_2^* - 1/T_2$ also changes and contributes a further dimensionless term $-\frac{|t+kT_R|}{T_2} S$ (since $\partial R'_2/\partial \log T_2 = 1/T_2$), again a ratio of times.) In linear coordinates the corresponding columns of the Jacobian differ by factors of order $T_1/B_1^+ \sim 10^3$ simply because of units, which says nothing about information.

2. The central symmetry is a straight line. The invariance (21) is the action of the multiplicative group $\kappa > 0$, $(M_0, B_1^+, T_1) \mapsto (\kappa^{-1}M_0, \kappa B_1^+, \kappa^{-2}T_1)$. In log coordinates it is a translation,

$$(\log M_0, \log B_1^+, \log T_1) \mapsto (\log M_0, \log B_1^+, \log T_1) + \log \kappa (-1, +1, -2), \quad (13)$$

so the near-null direction $v = (-1, 1, -2, 0, 0, 0)$ of Corollary 2 is the *same constant vector at every point* of parameter space, and the orbits are straight lines. In linear coordinates the orbits are curves $\{(M_0/\kappa, \kappa B_1^+, T_1/\kappa^2)\}$ whose tangent $(-M_0, B_1^+, -2T_1)$ changes from voxel to voxel, and the degeneracy can only be stated locally. The scaling-type structure of the problem is what makes log coordinates the ones in which it looks simplest.

3. The bounds become relative errors. With $D = \text{diag}(M_0, B_1^+, T_1, T_2, T_2^*, 1, 1)$, the Fisher information in linear coordinates is $\mathcal{I}_{\text{lin}} = D^{-1}\mathcal{I}D^{-1}$, hence

$$\sqrt{(\mathcal{I}^{-1})_{ll}} = \frac{\sqrt{(\mathcal{I}_{\text{lin}}^{-1})_{ll}}}{\theta_l^{\text{lin}}}, \quad (14)$$

which is the local (first-order) coefficient of variation. Relative error is the quantity of clinical interest (a 30 ms error matters at $T_1 = 300$ ms, not at 3000 ms) and is comparable across parameters with different units. For estimates, the delta method gives $\text{Var}(\log \hat{\theta}) \approx \text{Var}(\hat{\theta})/\theta^2$, again only to first order. For large spreads the two differ substantially: if $\log \hat{\theta}$ has standard deviation s and is Gaussian, the coefficient of variation of $\hat{\theta}$ is $\sqrt{e^{s^2} - 1}$, which is $\approx s$ for $s \lesssim 0.2$ but 9.1 for $s = 2.1$. We therefore call large values *log-parameter spreads* and read them as multiplicative factors: a spread s corresponds to a one-sigma interval $[\theta e^{-s}, \theta e^s]$.

4. The likelihood is closer to quadratic for the relaxation times. The CRLB and the asymptotic normality of maximum likelihood are second-order statements; they become accurate at lower SNR in coordinates where the log-likelihood is closer to quadratic over the noise-induced spread. We measured the asymmetry of the noise-free cost $\|s(\theta) - s(\theta_0)\|^2$ for equal steps up and down in each coordinate, holding the others fixed (reference tissue; a ratio of 1 means symmetric):

Parameter	$\pm 20\%$		$\pm 40\%$	
	linear	log	linear	log
T_1	0.58	0.87	0.32	0.75
T_2	0.78	1.16	0.60	1.34
B_1^+	8.2	8.3	1.85	1.82
M_0	1.00	1.49	1.00	2.23

The log coordinates are clearly better for T_1 , which has the largest bound and hence the largest excursions, and somewhat better for T_2 . For B_1^+ the asymmetry comes from the 360° structure of scan 2 and is the same in both coordinates. For M_0 linear coordinates are exactly quadratic, because the signal is linear in M_0 . So argument 4 does not justify $\log M_0$; arguments 2 and 3 do. Consistently, estimates of relaxation times are right-skewed in linear units (for example $T_1 = T_R/\ln(\cdot)$ in Eq. 16 has a heavy right tail), and we analyse Monte Carlo errors on $\log \hat{\theta}$, reporting standard deviation, bias, RMSE and a robust spread side by side (Section 5).

5. Optimisation. Log coordinates make positivity automatic. A step of size 0.1 is a 10% change whatever the magnitude of the parameter, and the Jacobian columns have comparable norms, so the damped Gauss–Newton steps are well scaled from the first iteration. Positivity is not the whole physical domain, however: see the ordering constraint below.

What does not depend on the choice. The change from linear to log coordinates has a diagonal Jacobian D . Therefore the correlation matrix of $\mathcal{I}^{-1}$, the normalised information $\tilde{\mathcal{I}}$ (and its eigenvalues and condition number), the variance inflation factors, and every statement of the form “parameter l is (not) determined” are *identical* in both parameterisations at a given point. The conclusions of Sections 3–5 do not rest on this choice. What the logarithm adds is a global picture (straight symmetry orbits), unit-free and relative error measures, better-behaved statistics, and easier optimisation.

Which parameters are not logged. $\Delta\omega$ is a signed frequency and ϕ_0 a periodic phase: neither is a scale parameter, and both enter (8) linearly in the exponent, so they stay linear. For the reversible decay we use $\log T_2^*$ rather than $\log R_2'$: R_2' can be arbitrarily close to zero (here $R_2' = 1/60 - 1/70 \text{ ms}^{-1}$), and a relative error on a near-zero rate is large and uninformative. In the same coordinates the bound on $\log R_2'$ is 366 per unit σ/M_0 , against 47 for $\log T_2^*$, for exactly the same information.

The ordering constraint. The Lorentzian model requires $R_2' = 1/T_2^* - 1/T_2 \geq 0$, i.e. $T_2^* \leq T_2$. Independent $\log T_2$ and $\log T_2^*$ coordinates enforce positivity of each but not this ordering. The CRLB is computed at an interior point ($R_2' > 0$) and is unaffected, but an estimator must enforce the ordering explicitly; our maximum-likelihood fit projects onto $\log T_2^* \leq \log T_2$ (Section 5.1) and reports how often the constraint is active. When it is active the estimator is no longer (locally) unbiased and the comparison with the CRLB loses its footing for those voxels.

3 The fundamental limitation: a scaling invariance

3.1 Two invariants per scan

The steady state of one scan depends on the flip angle and T_1 only through two combinations. This exact result underlies everything in this section and in Section 6.

Theorem 1 (Exact two-invariant reduction). *Let $\varphi = 0$ and define, for one scan with flip angle α ,*

$$u = \frac{E_1 - \cos \alpha}{1 - E_1 \cos \alpha}, \quad v = \frac{\sin \alpha (1 - E_1)}{1 - E_1 \cos \alpha}. \quad (15)$$

Then for every pathway k and every off-resonance $\Delta\omega$,

$$a_k(\alpha, T_1, T_2, \Delta\omega) = v G_k(u, E_2, \Delta\omega) \quad (16)$$

for functions G_k that do not depend on α or T_1 otherwise. The magnitude of v is itself a function of u and T_1 :

$$|v| = \sqrt{1 - u^2} \tanh\left(\frac{T_R}{2T_1}\right). \quad (17)$$

Proof. Write the pre-pulse state of one isochromat as (x, y, z) and $w = x + iy$. The pulse $R_x(\alpha)$ maps it to $(x, cy - sz, sy + cz)$ with $c = \cos \alpha$, $s = \sin \alpha$; relaxation and precession by $\beta = \psi + \Delta\omega T_R$ then give the next pre-pulse state. The steady-state equations are

$$w = E_2 e^{i\beta} (x + i(cy - sz)), \quad z = E_1 (sy + cz) + 1 - E_1. \quad (18)$$

The second gives $z = (E_1 sy + 1 - E_1)/(1 - E_1 c)$. Substituting,

$$cy - sz = y \frac{c - E_1 c^2 - E_1 s^2}{1 - E_1 c} - \frac{s(1 - E_1)}{1 - E_1 c} = -u y - v.$$

The post-pulse transverse magnetisation is therefore $m_\perp = x - iuy - iv$ with $w = E_2 e^{i\beta} m_\perp$, i.e.

$$m_\perp = \text{Re}(E_2 e^{i\beta} m_\perp) - iu \text{Im}(E_2 e^{i\beta} m_\perp) - iv.$$

This is a real-linear equation for $m_\perp$ whose coefficients involve only u , E_2 and β , with inhomogeneous term $-iv$. Hence $m_\perp = v g(u, E_2, \beta)$, and its Fourier coefficients over ψ have the form (16). For (17), use $1 \mp u = (1 \mp E_1)(1 \pm c)/(1 - E_1 c)$, so $1 - u^2 = (1 - E_1^2)s^2/(1 - E_1 c)^2$ and $v^2 = (1 - u^2)(1 - E_1)/(1 + E_1)$, with $(1 - E_1)/(1 + E_1) = \tanh(T_R/2T_1)$. $\square$

Corollary 1 (One scan cannot separate M_0, B_1^+, T_1). *For a single flip angle, the signal depends on (M_0, B_1^+, T_1) only through $(u, M_0 v)$. The level sets of these two functions are curves in the three-dimensional parameter space, so the Fisher information about (M_0, B_1^+, T_1) from one scan is exactly singular, at every flip angle.*

Numerically, for one scan at 15° the amplitudes of six pathways are reproduced to relative precision 10^{-15} by $(\kappa\alpha, T_1', M_0')$ for every κ tested (1.05 to 2), and the scan-1-only bound on $\log T_1$ with B_1^+ unknown is 2.4×10^8 (Section 4): singular to numerical precision. In the small-angle limit, $u \rightarrow (\alpha^2/2 - \rho)/(\alpha^2/2 + \rho)$ and $v \rightarrow \alpha\rho/(\rho + \alpha^2/2)$ depend on α and ρ only through α^2/ρ and an overall factor α , which is the scaling of Proposition 2 below. The two-scan problem is the question of when two such reductions, tied together by a common B_1^+ , have a unique joint solution.

Proposition 2 (Small-angle scaling invariance). *Let $\rho = T_R/T_1$, $\varphi = 0$, and fix T_R, T_2 (hence $E_2 < 1$) and a pathway order k . In the joint limit $\alpha \rightarrow 0$, $\rho \rightarrow 0$ with $c = \alpha^2/(2\rho)$ held fixed,*

$$a_k(\alpha, T_1) = \alpha f_k(c, E_2) + O(\alpha^3 + \alpha\rho), \quad (19)$$

with f_k given in (26) below. Consequently, for any fixed $\kappa > 0$,

$$M_0 a_k(\alpha, T_1) - \frac{M_0}{\kappa} a_k(\kappa\alpha, T_1/\kappa^2) = O(\alpha^3 + \alpha\rho), \quad (20)$$

i.e. the signal (8) of every scan with a small flip angle, in every pathway and echo, is invariant to leading order under

$$(M_0, B_1^+, T_1) \mapsto (M_0/\kappa, \kappa B_1^+, T_1/\kappa^2). \quad (21)$$

The remainder is additive (safe where f_k is small) and is not uniform in κ : the limit is taken with κ fixed.

Proof. Work with one isochromat, $\Delta\omega = 0$, and write $m_\perp = m_x + im_y$ and m_z for the state just after the pulse. Free precession and relaxation give the pre-pulse state $m_\perp^- = E_2 e^{i\psi} m_\perp$, $Z \equiv m_z^- = E_1 m_z + (1 - E_1)$. The pulse $R_x(\alpha)$ gives

$$m_\perp = m_\perp^- - i \sin \alpha Z + i(\cos \alpha - 1) \text{Im } m_\perp^-, \quad m_z = \sin \alpha \text{Im } m_\perp^- + \cos \alpha Z. \quad (22)$$

In the steady state $Z = O(1)$ and $m_\perp = O(\alpha)$, so the last term in $m_\perp$ is $O(\alpha^3)$ and $\sin \alpha = \alpha + O(\alpha^3)$. Solving the transverse equation,

$$m_\perp = \frac{-i\alpha Z}{1 - E_2 e^{i\psi}} + O(\alpha^3). \quad (23)$$

The denominator is bounded away from zero, $|1 - E_2 e^{i\psi}| \geq 1 - E_2 > 0$, uniformly in ψ . For the longitudinal equation use $E_1 = 1 - \rho + O(\rho^2)$, so $m_z = Z - \rho(1 - Z) + O(\rho^2)$, and $\cos \alpha = 1 - \alpha^2/2 + O(\alpha^4)$:

$$\rho(1 - Z) = -\alpha \operatorname{Im} m_\perp^- + \frac{\alpha^2}{2} Z + O(\alpha^4 + \rho^2 + \alpha^2 \rho). \quad (24)$$

From (23), $\operatorname{Im} m_\perp^- = -\alpha Z \operatorname{Re}[E_2 e^{i\psi}/(1 - E_2 e^{i\psi})] = -\alpha Z (E_2 \cos \psi - E_2^2)/|1 - E_2 e^{i\psi}|^2$, and

$$\frac{E_2 \cos \psi - E_2^2}{|1 - E_2 e^{i\psi}|^2} + \frac{1}{2} = \frac{1}{2} H(\psi), \quad H(\psi) = \frac{1 - E_2^2}{|1 - E_2 e^{i\psi}|^2} = \frac{1 - E_2^2}{1 - 2E_2 \cos \psi + E_2^2}.$$

Hence $\rho(1 - Z) = \frac{\alpha^2}{2} H(\psi) Z$ to leading order, i.e. $Z = [1 + c H(\psi)]^{-1}$ with $c = \alpha^2/(2\rho)$, and

$$m_\perp(\psi) = \frac{-i\alpha}{(1 - E_2 e^{i\psi}) [1 + c H(\psi)]} + O(\alpha^3 + \alpha\rho), \quad (25)$$

where $1 + cH \geq 1$ keeps the expression uniformly bounded, so the remainder is uniform in ψ . The pathway amplitude is the Fourier coefficient

$$a_k = \frac{1}{2\pi} \int_0^{2\pi} m_\perp(\psi) e^{-ik\psi} d\psi = \alpha f_k(c, E_2) + O(\alpha^3 + \alpha\rho),$$

$$f_k(c, E_2) = \frac{1}{2\pi} \int_0^{2\pi} \frac{-i e^{-ik\psi} d\psi}{(1 - E_2 e^{i\psi}) [1 + c H(\psi)]}, \quad (26)$$

and a uniform remainder in ψ bounds the remainder of every Fourier coefficient. Under $\alpha \rightarrow \kappa\alpha$, $\rho \rightarrow \kappa^2\rho$, c is unchanged and the leading term scales by κ , which $M_0 \rightarrow M_0/\kappa$ compensates. Since B_1^+ multiplies every flip angle (1), $\alpha \rightarrow \kappa\alpha$ is $B_1^+ \rightarrow \kappa B_1^+$. The echo factors in (8) do not involve α or T_1 . $\square$

Remark 2 (Checks). For $k = 0$ and $k = -1$ the same conclusion follows directly from the closed forms (7): to leading order $E_1 - \cos \alpha \simeq \alpha^2/2 - \rho$, $1 - E_1 \cos \alpha \simeq \rho + \alpha^2/2$ and $q \simeq 2E_2\rho$ are homogeneous of degree one in (ρ, α^2) , so the brackets depend only on c , and $\tan(\alpha/2) \simeq \alpha/2$. Numerically, with $c = 0.75$, the leading-order formula (26) matches the exact solver (6) for all pathways $k = -3, \dots, 2$ with maximum error (relative to $|a_0|$) 2.5×10^{-3} , 6.2×10^{-4} , 1.5×10^{-4} and 3.8×10^{-5} at $\alpha = 15^\circ, 7.5^\circ, 3.75^\circ, 1.875^\circ$, the factor of four per halving expected from an $O(\alpha^2)$ relative remainder; the scaling defect for $\kappa = 2$ falls at the same rate. On the full model with all three pathways and the $15^\circ/30^\circ$ protocol, fixing B_1^+ wrong by ε shifts the best fit by $\delta \log T_1 = -2.06 \varepsilon$ and $\delta \log M_0 = -1.02 \varepsilon$, against the predicted -2 and -1 .

Corollary 2 (Leading-order degeneracy). *Let $v = (-1, +1, -2, 0, 0, 0, 0)$ in the coordinates of θ . If all flip angles are small, $\partial_v S_{ijk}$ vanishes at leading order (it is $O(\alpha^3 + \alpha\rho)$ while the other directional derivatives are $O(\alpha)$), so $v^\top \mathcal{I} v$ is small relative to the diagonal of $\mathcal{I}$ and $\mathcal{I}$ is ill-conditioned. Small-angle acquisitions, however many and however their angles differ, cannot remove this leading-order degeneracy, because B_1^+ scales all of them by the same factor.*

Remark 3 (What the corollary does and does not say). The degeneracy is only of leading order. At finite angles the higher-order terms break it, $\mathcal{I}$ is nonsingular, and the parameters are identifiable in principle; the corollary says that separating them is extremely noise-sensitive. For the reference tissue the condition number of $\tilde{\mathcal{I}}$ is 2.6×10^4 for $30^\circ/60^\circ$, 5.0×10^5 for $15^\circ/30^\circ$ and 2.5×10^7 for $3.75^\circ/7.5^\circ$. Repeating acquisitions adds information in the usual way (all bounds shrink as $1/\sqrt{N}$), so a sufficiently high SNR resolves the poorly determined direction; it does not improve the conditioning. The bounds apply to unbiased estimators. A biased estimator (a prior, regularisation, or a learned network) can report a smaller spread along v , but that reduction comes from the prior, not from the data.

3.2 How the published protocol escapes

Proposition 3 (Periodicity). *$|a_k(\alpha)|$ is even and 2π -periodic in α , and $a_k(-\alpha) = -a_k(\alpha)$ for all k (a global sign). Hence two protocols whose actual flip angles agree up to sign and multiples of 2π in every scan carry identical Fisher information about every parameter when B_1^+ is known. In particular, at $B_1^+ = 1$ the protocols $15^\circ/330^\circ$ and $15^\circ/30^\circ$ are equivalent with B_1^+ known.*

Proof. $R_x(-\alpha) = R_z(\pi)R_x(\alpha)R_z(-\pi)$, i.e. RF phase $\varphi = \pi$. Since precession and relaxation commute with R_z , the steady state for phase π is $R_z(\pi)$ applied to the one for phase 0, which multiplies every F_k by -1 . Periodicity is that of R_x . A known, scan-wide sign does not change $\mathcal{I}$, and with B_1^+ known the actual angles are fixed. $\square$

The equivalence is a statement about actual angles, not nominal ones: $330^\circ B_1^+$ and $-30^\circ B_1^+$ agree modulo 360° only when $360^\circ B_1^+ \equiv 0$, i.e. at $B_1^+ = 1$ within the physical range. At $B_1^+ = 1.1$, for example, the second angles are 363° and 33° , and the known- B_1^+ T_1 bounds are 29.6 and 29.0 per unit σ/M_0 . Near $B_1^+ = 1$ the information about T_1 given B_1^+ is therefore nearly the same for both protocols, and the value of the large pulse lies in the B_1^+ derivative. The effective angle of scan 2 is $\alpha_2^{\text{eff}} = 330^\circ B_1^+ - 360^\circ$, so

$$\frac{\partial \log |\alpha_2^{\text{eff}}|}{\partial \log B_1^+} = \frac{330 B_1^+}{330 B_1^+ - 360}, \quad \text{whereas} \quad \frac{\partial \log \alpha_1}{\partial \log B_1^+} = +1. \quad (27)$$

At $B_1^+ = 1$ the first ratio is -11 ; it is -2.75 at $B_1^+ = 0.8$ and diverges as $\alpha_2^{\text{eff}} \rightarrow 360^\circ$. The invariance (21) requires all angles to scale by the same factor; under a B_1^+ change the second angle moves much faster and in the opposite direction, so the near-null direction v is lifted. The near- 360° pulse acts as a built-in B_1^+ measurement.

3.3 Global aliases

The local analysis says nothing about *distant* parameter values that produce the same data. Theorem 1 makes them easy to find. For given T_2 , the amplitudes of scan i determine u_i and $M_0 v_i$. For a trial value of E_1 , inverting the Möbius map (15) gives

$$\cos \alpha_i = \frac{E_1 - u_i}{1 - u_i E_1}, \quad (28)$$

and B_1^+ must satisfy $B_1^+ \alpha_i^{\text{nom}} = \pm \arccos(\cos \alpha_i) + 2\pi m_i$ for both scans at once. Each choice of sign and of m_i is a candidate branch; the true parameters are one solution, and other branches generally admit exact solutions of their own. Which of them the data can exclude depends on what the data retain:

- *Magnitudes* retain u_i and $M_0|v_i|$, i.e. $\cos \alpha_i$ but not the sign of $\sin \alpha_i$. The branch $\alpha_2 = 360^\circ + x$ is then indistinguishable from the true $\alpha_2 = 360^\circ - x$ (Section 6).
- *Complex data* with a phase shared by both scans also retain the relative sign of v_1 and v_2 , i.e. of $\sin \alpha_1 \sin \alpha_2$. This removes the $360^\circ + x$ branch. It cannot remove branches with the same sign pattern that differ by 2π in α_2 , such as $\alpha_2 = 720^\circ - x'$.

Table 1 lists the aliases found by least squares restricted to each branch, on noise-free data. All fit to machine precision.

Table 1: Exact aliases on noise-free data (relative residual below 10^{-14} in every case). Magnitude alias: best magnitude fit with the opposite FID sign (α_2 on the other side of 360°). Complex alias: best complex fit with $\alpha_2 \geq 540^\circ$.

Truth		Magnitude alias		Complex alias	
B_1^+	T_1 (ms)	B_1^+	T_1 (ms)	B_1^+	T_1 (ms)
1.00	1500	1.199	1038	2.008	359
0.90	1200	1.350	527	1.865	271
1.05	850	1.135	726	2.100	204
0.80	2500	1.480	717	1.769	497
1.08	1000	1.102	960	2.160	240

The complex aliases need $B_1^+ \approx 1.8$ – 2.2 and are excluded by the assumption, stated in [1], that $\alpha_2^{\text{nom}} B_1^+ \in [180^\circ, 540^\circ]$, i.e. $B_1^+ < 540/330 \approx 1.64$ for this protocol. The magnitude aliases lie at $B_1^+ \approx 1.1$ – 1.5 , inside the physiological range, so no range assumption removes them. This is why [1] uses the relative phase of the two FID images to choose the branch. As $\alpha_2^{\text{eff}} \rightarrow 360^\circ$ the true and alias solutions merge (last row): the branch question then becomes moot, but also unanswerable, since the scan-2 FID vanishes. Being a global property, none of this is visible in the local CRLB; locally, B_1^+ is determined *best* near 360° (Fig. 4).

4 Cramér–Rao bounds and conditioning

Unless stated otherwise the tissue is the reference used for protocol optimisation in [1]: $T_1/T_2/T_2^* = 1500/70/60$ ms, $B_1^+ = 1$.

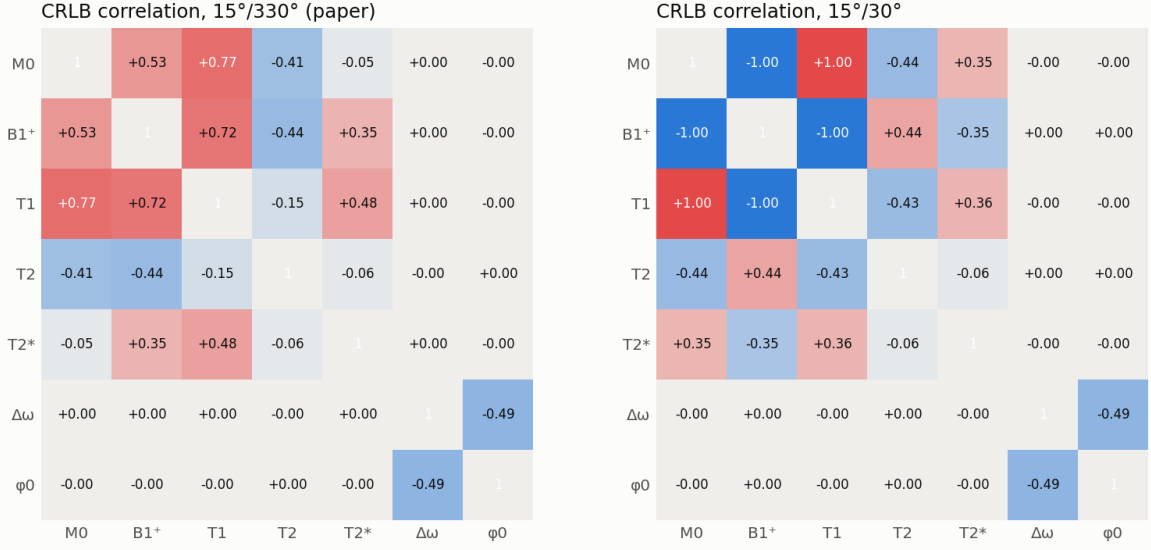
Table 2: CRLB per unit σ/M_0 : bound on the standard deviation of $\log M_0$, $\log B_1^+$, $\log T_1$, $\log T_2$, $\log T_2^*$; rad/ms for $\Delta\omega$. Multiply by σ/M_0 for a given SNR; the result approximates a relative standard deviation only when it is small (Section 2.1).

Protocol	Tissue ($T_1/T_2/T_2^*$, ms)	M_0	B_1^+	T_1	T_2	T_2^*	$\Delta\omega$
15°/330°	WM (850/66/50)	12.5	2.21	27.3	16.0	27.2	0.30
15°/330°	GM (1350/90/60)	14.7	2.75	33.1	20.7	38.2	0.31
15°/330°	reference (1500/70/60)	19.0	2.91	40.5	23.1	47.3	0.37
15°/330°	reference, B_1^+ known	16.1	–	28.0	20.7	44.2	0.37
15°/30°	reference	1050	1020	2110	23.1	47.3	0.37
15°/30°	reference, B_1^+ known	16.1	–	28.0	20.7	44.2	0.37

Table 2 shows the two effects at once. With 15°/30° the M_0 – B_1^+ – T_1 block is essentially singular while T_2 , T_2^* and $\Delta\omega$ are untouched; with B_1^+ known the two protocols coincide (Proposition 3).

Table 3: Conditioning of the information matrix (reference tissue).

	15°/330°	15°/30°
Eigenvalues of $\tilde{\mathcal{I}}$	0.024, 0.11, 0.22, 0.51, 0.52, 1.49, 4.13	7.7×10^{-6} , 0.11, 0.29, 0.51, 0.76, 1.49, 3.84
Condition number $\kappa(\tilde{\mathcal{I}})$	173	5.0×10^5
Variance inflation $M_0/B_1^+/T_1$	16.7/4.6/26.6	51 000/7 700/72 000
Correlations (B_1^+, T_1) , (B_1^+, M_0) , (T_1, M_0)	0.72, 0.53, 0.77	-0.9999, -0.9999, 0.9999
$\partial \log T_1 / \partial \log B_1^+$, B_1^+ fixed, all data	+10.1	-2.06
$\partial \log T_1 / \partial \log B_1^+$, B_1^+ fixed, scan 1 only	-2.02	-2.02

Figure 1: Correlation matrices of the CRLB covariance $\mathcal{I}^{-1}$. Right: the M_0 – B_1^+ – T_1 block of the small-angle protocol is perfectly correlated, the signature of the invariance (21).

Where the remaining T_1 uncertainty comes from. Freeing parameters one group at a time gives the hierarchy

Unknown parameters	T_1 bound (per unit σ/M_0)
T_1 only	7.9
+ T_2 , T_2^* , $\Delta\omega$, ϕ_0	21.1
+ M_0	28.0
+ B_1^+ (15°/330°)	40.5
+ B_1^+ (15°/30°)	2110

With the published protocol, freeing B_1^+ last costs a factor 1.45; freeing the transverse relaxation times costs 2.7 and M_0 1.3. (These factors depend on the order in which the groups are freed; the variance inflation factors of Table 3 are the order-free summary.) At 15° and $T_R = 25$ ms ($T_1 \gg T_R$), T_1 is weakly encoded in the steady-state amplitudes. Consistently, the weakest direction of the paper protocol in log coordinates (Fig. 2) is dominated by T_1 (component -0.85), with M_0 and T_2^* (-0.36 each) and T_2 (+0.15); its B_1^+ component is only -0.05. With this protocol B_1^+ is no longer the main source of T_1 uncertainty.

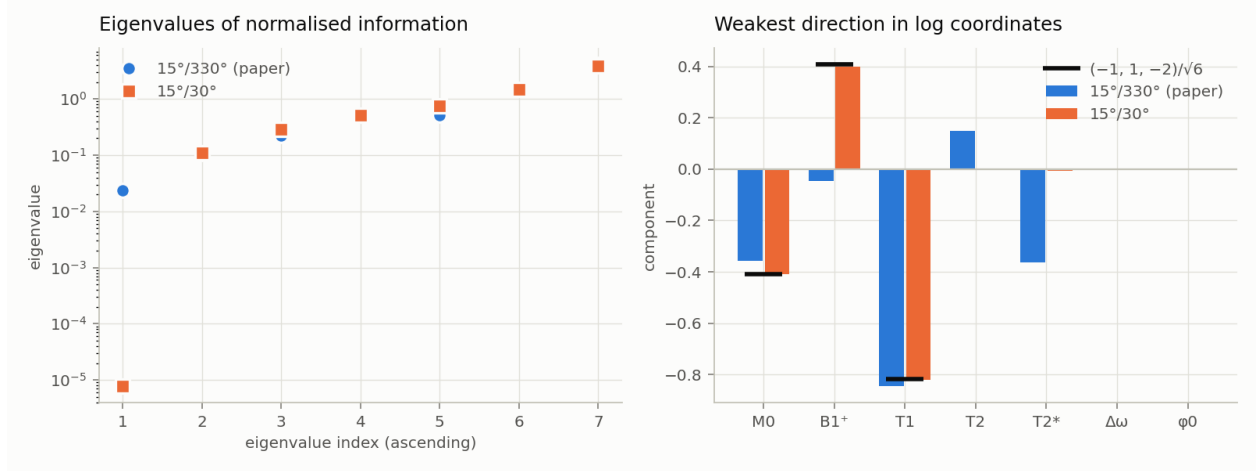


Figure 2: Left: eigenvalues of the normalised information $\tilde{\mathcal{I}}$. Right: the weakest direction mapped back to the original log coordinates, $w \propto D^{-1/2}u$ with u the eigenvector of the smallest eigenvalue of $\tilde{\mathcal{I}}$ (normalised to unit length, sign fixed by the T_1 component). For $15^\circ/30^\circ$ it coincides with the symmetry direction $(-1, 1, -2)/\sqrt{6}$ (black marks). For $15^\circ/330^\circ$ the smallest eigenvalue is three thousand times larger and the weakest direction is mostly T_1 , with M_0 and T_2^* ; its B_1^+ component is only -0.05 .

Which measurements carry the information. Separating the data used from whether B_1^+ is known gives the T_1 bounds

Data	B_1^+ unknown	B_1^+ known
both scans (full resolution)	40.5	28.0
scan 1 only	2.4×10^8	57.1

Scan 1 alone cannot determine B_1^+ and T_1 jointly at all: by Corollary 1 the one-scan information is exactly singular, whatever the flip angle. With B_1^+ known, adding the high-flip scan still halves the bound on the standard deviation of $\log T_1$ (57.1 to 28.0), because scan 2 carries T_1 information of its own.

Protocol dependence. Fig. 3 shows that the T_1 bound with B_1^+ unknown falls steeply with α_2^{nom} and, from about 210° , stays within $1.5\times$ of the B_1^+ -known bound. Over (α_1, α_2) the optimum is $10^\circ/340^\circ$ (bound 32.9), only 19% better than $15^\circ/330^\circ$ (Fig. 5). Across a 3 T-typical B_1^+ range 0.7–1.3 the T_1 bound stays between 40 and 65, and it grows with T_1 itself (15 at 300 ms, 100 at 4000 ms; Fig. 4). Removing the $k = +1$ pathway raises the T_1 bound from 40.5 to 67; removing the echo pathway $k = -1$ raises it to 146 and the T_2 bound from 23 to 208.

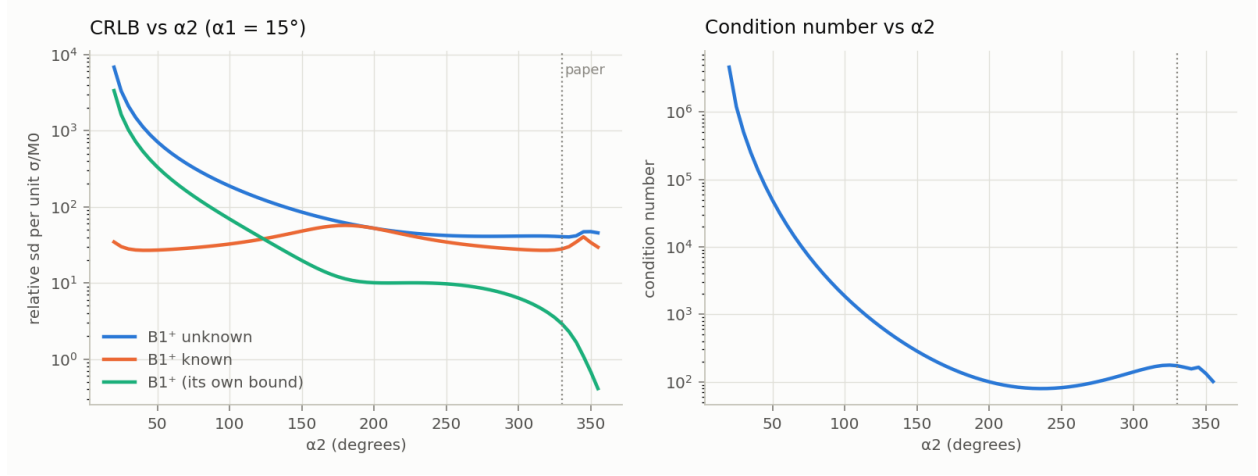


Figure 3: Bounds and condition number versus the nominal second flip angle ($\alpha_1^{\text{nom}} = 15^\circ$).

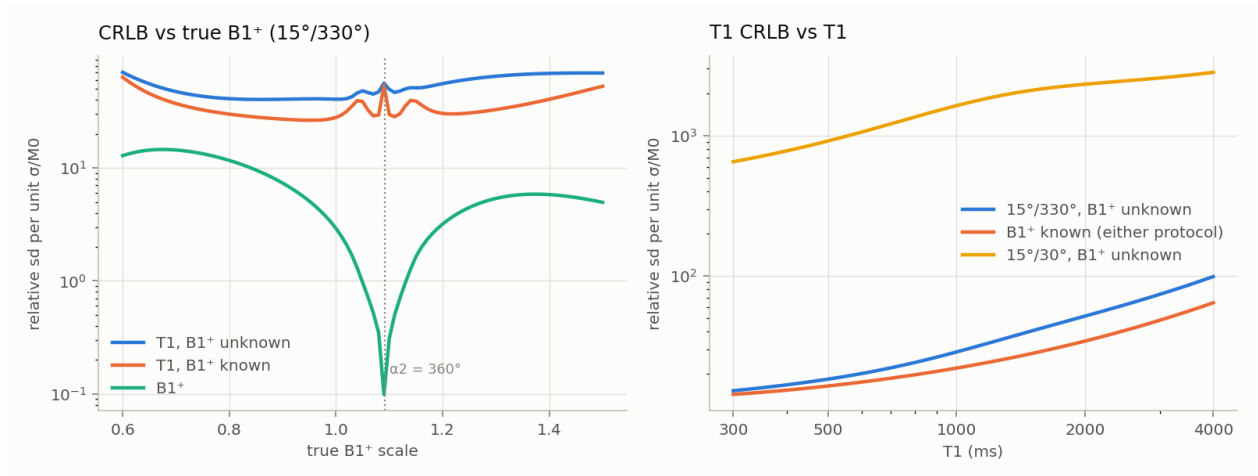


Figure 4: Left: bounds versus the true B_1^+ for the published protocol; B_1^+ is best determined where $\alpha_2^{\text{eff}} = 360^\circ$. Right: T_1 bound versus T_1 .

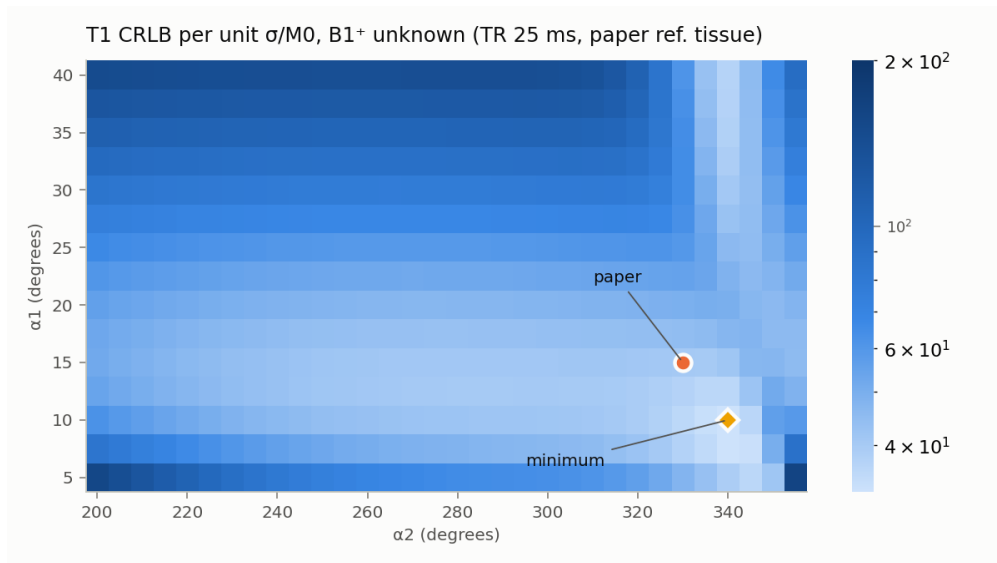


Figure 5: T_1 bound with B_1^+ unknown over nominal (α_1, α_2) .

5 Forward model or inverse?

The CRLB characterises the forward model and protocol; it says nothing about whether a particular reconstruction attains it. To separate the two we compare three estimators of the *same* model on simulated data:

1. the **analytic inverse** of [1], implemented equation by equation: $\Delta\omega$ from echo phases; R_2 , R'_2 from a log-linear fit of their Eq. 1 to scan 1; real-valued F states (Eq. 18); the mixing factor (Eqs. 10–11); the flip angle from Eq. 15; T_1 and M_0 from Eqs. 16–17 using scan 1. We also run it with B_1^+ given, mimicking the paper’s low-resolution, polynomial-smoothed B_1^+ map;
2. a **model-fit baseline**: the same sequential structure, but with B_1^+ , T_1 , M_0 fitted by least squares to the FID and echo amplitudes of both scans;
3. a **joint maximum-likelihood (ML) fit** of (8) to all complex data (Section 5.1).

All three are exact on noise-free data (relative errors below 10^{-11} for the analytic inverse). Table 4 reports robust standard deviations (interquartile range/1.349 of the log-estimates) over 1000 noise realisations.

5.1 The joint maximum-likelihood estimator

Objective. Stack the real and imaginary parts of all measured voxel signals ($2 \text{ scans} \times 3 \text{ pathways} \times 3 \text{ echoes}$) into $y \in \mathbb{R}^{36}$ and the model (8) into $s(\theta) \in \mathbb{R}^{36}$, with θ as in Section 2. Under independent Gaussian noise of variance σ^2 per component the negative log-likelihood is, up to constants,

$$-\log p(y \mid \theta) = \frac{1}{2\sigma^2} \|r(\theta)\|^2, \quad r(\theta) = s(\theta) - y, \quad (29)$$

so the ML estimate is the nonlinear least-squares solution $\hat{\theta} = \arg \min_{\theta} \|r(\theta)\|^2$. It does not depend on σ . The model $s(\theta)$ is evaluated with the exact steady-state solver (6) (256 isochromats), so the estimator uses the same forward model as the CRLB, including all three pathways, both scans, and the phase.

Parameterisation and constraints. The five positive parameters are optimised in log coordinates (Section 2.1); $\Delta\omega$ and ϕ_0 are unconstrained. The fit is a *projected* Levenberg–Marquardt iteration: after every step the parameters are projected onto

- the physical domain of the Lorentzian model, $\log T_2^* \leq \log T_2$ ($R'_2 \geq 0$);
- a safeguard box $B_1^+ \in [e^{-1}, e]$, $T_1 \in [10, 20\,000] \text{ ms}$, $T_2 \in [1, 5000] \text{ ms}$, which excludes far-unphysical steps (at which the steady-state system can become singular).

For every voxel we record whether a constraint is active at the solution. Such estimates are constrained ML estimates: they are not locally unbiased and are not covered by the interior CRLB.

Initialisation. The magnitude parameters and $\Delta\omega$ are taken from a sequential estimator: the model-fit baseline, or the analytic inverse where its output is valid. The global phase, which the

sequential estimators do not provide, has a closed-form optimum for fixed remaining parameters. With $S_{ijk}^{(0)}$ the model evaluated at the initial values and $\phi_0 = 0$,

$$\phi_0^{(0)} = \arg\left(\sum_{i,j,k} S_{ijk} \overline{S_{ijk}^{(0)}}\right), \quad (30)$$

which minimises $\|r\|^2$ over ϕ_0 alone.

Levenberg–Marquardt iteration. Each voxel is fitted independently; all voxels are processed in parallel as a batch. At iteration n , with $J = \partial r / \partial \theta \in \mathbb{R}^{36 \times 7}$:

1. *Jacobian.* Forward differences, $J_{:,l} = [r(\theta + h e_l) - r(\theta)]/h$ with $h = 10^{-6}$ (all quantities in double precision).
2. *Damped Gauss–Newton step* with Marquardt’s diagonal scaling,

$$(J^\top J + \lambda \text{diag}(J^\top J)) \Delta = -J^\top r, \quad \theta_{\text{new}} = \theta + \Delta. \quad (31)$$

The diagonal scaling makes the step invariant to rescaling individual parameters, which matters here because the information matrix spans several orders of magnitude.

3. *Acceptance.* If $\|r(\theta_{\text{new}})\|^2 < \|r(\theta)\|^2$ the step is accepted and $\lambda \leftarrow 0.3\lambda$ (towards Gauss–Newton); otherwise it is rejected and $\lambda \leftarrow 10\lambda$ (towards scaled gradient descent). λ starts at 10^{-2} and is kept in $[10^{-9}, 10^9]$.

We run a fixed 40 iterations. Near the optimum $\lambda \rightarrow 0$ and the iteration becomes Gauss–Newton. For a zero-residual problem (noise-free data) and a nonsingular Jacobian, Gauss–Newton converges locally quadratically; in noise-free tests the parameters are recovered to 10^{-15} relative error from starts perturbed by 5%. With noise the residual at the solution is nonzero and the local convergence of Gauss–Newton is in general only linear, with a rate governed by the residual size and the curvature of the model; 40 iterations were ample in our experiments (the cost no longer decreases).

Multiple starts. Nonlinear least squares can converge to a local minimum, in particular across the 360° branch of the second flip angle or, for the small-angle protocol, anywhere along the nearly flat symmetry valley (21). We therefore run the iteration from four starts and keep, per voxel, the solution with the lowest final cost: the model-fit estimate; for $15^\circ/330^\circ$ the analytic-inverse estimate (where valid), for $15^\circ/30^\circ$ the model-fit estimate moved along the symmetry direction ($B_1^+ \times 1.1$, $T_1/1.21$, $M_0/1.1$); and the model-fit estimate with T_1 replaced by 700 ms and by 2000 ms (and $T_2^* \leq 0.85 T_2$).

In simulation the global optimum can be checked: if the final cost exceeds the cost at the true parameters, the optimiser has certainly not found the maximum likelihood. With only the first two starts this happened in 19 of 1000 voxels at $\text{SNR}(M_0) = 300$ (spurious minima near $T_1 \approx 6000\text{--}10\,000$ ms with $R_2' = 0$), which inflated the standard deviation of $\log \hat{T}_1$ from 0.13 to 0.26 and produced the heavy tails noted by a reviewer of an earlier version. With four starts no voxel failed this check (Table 6).

When the ML fit should attain the bound. At the solution, the Gauss–Newton approximation of the covariance is $\sigma^2(J^\top J)^{-1}$, and $J^\top J/\sigma^2$ evaluated at the true parameters is exactly the Fisher information of Section 2. Under standard regularity conditions (interior true parameter, identifiability,

smoothness) the unconstrained ML estimator is consistent and asymptotically efficient: as $\text{SNR} \rightarrow \infty$ its covariance converges to $\mathcal{I}^{-1}$ and its bias vanishes. At finite SNR none of this is guaranteed. The estimator can be biased, it can have heavy tails, and when constraints are active it can even have a variance *below* the bound, because it is then no longer unbiased. A meaningful efficiency comparison therefore needs (i) the ordinary standard deviation, not only a robust spread, (ii) the bias and RMSE, and (iii) the fraction of estimates on a constraint; Table 4 reports all of them.

Cost. Each iteration needs eight model evaluations (one plus seven finite-difference directions). For 10^3 voxels the full fit takes about a minute on a laptop CPU. That is fast enough for simulation studies, but slow for whole-brain data ($\sim 5 \times 10^6$ voxels), which is one motivation for the learned estimators in the project plan.

Reading Tables 4–6. All ratios below use ordinary standard deviations (SD); robust spreads are listed alongside because they can hide heavy tails. The Monte Carlo standard error of an SD estimated from 1000 samples is about 2%.

- **The bound is attained where its assumptions hold.** For $15^\circ/330^\circ$ at $\text{SNR}(M_0) = 1000$ and 3000 the joint ML fit has SD/CRLB between 0.99 and 1.02 for T_1 , B_1^+ , T_2 and M_0 , negligible bias ($|\text{mean}| < 10^{-3}$ in log units), no constrained estimates (at most 0.3%) and no optimiser failures. In this regime the forward model *contains* the information that Table 2 promises, and an estimator extracts it; the published protocol can separate B_1^+ and T_1 .
- **At SNR 300 the comparison is only approximate.** The ML SD/CRLB is 1.00 for B_1^+ and 0.95 for T_1 , but the constraint $R_2' \geq 0$ is active in 19.5% of voxels (true R_2' is small: $T_2^* = 60$ vs $T_2 = 70$ ms). Those estimates are not unbiased, which is how the T_1 SD can fall slightly below the bound. The bias is small (-0.007 in $\log T_1$).
- **The analytic inverse is well above the bound.** Its T_1 SD is $3.2\times$ the bound at SNR 3000, $5.1\times$ at SNR 1000 and $7.1\times$ at SNR 300, where it is also strongly biased ($+0.44$ in $\log T_1$, a factor 1.56) and 1.9% of voxels have no valid T_1 (Eq. 16 needs $(Xc - 1)/(X - c) > 1$). Its B_1^+ SD is $4.8\times$ the bound at SNR 3000 but $23\times$ and $28\times$ at SNR 1000 and 300, with heavy tails (SD much larger than the robust spread at SNR 1000). The model-fit baseline, which shares the sequential structure but not the closed forms, is similar (T_1 : $2.8\times$, $3.9\times$, $7.0\times$): the loss comes mainly from the sequential structure.
- **With B_1^+ given, compare with the matched bound.** The analytic inverse computes T_1 from scan 1 with B_1^+ fixed, so its matched bound is the scan-1, B_1^+ -known bound of 57.1 per unit σ/M_0 . Its SD is then $1.9\times$ that bound at SNR 1000 and 3000 and $2.8\times$ at SNR 300 (Table 6). Against the two-scan, B_1^+ -known bound of 28.0 the ratios are 3.8–5.7; against the B_1^+ -unknown bound of 40.5 (the reference used in an earlier version of this note) the comparison is not like-for-like.
- **The small-angle control does not support an efficiency claim.** For $15^\circ/30^\circ$ the safeguard box is active in 35% (SNR 1000) and 10% (SNR 3000) of ML estimates, which are biased ($+0.41$ and $+0.24$ in $\log T_1$). At SNR 1000 the ML SD is therefore *below* the bound (0.65 of it), and at SNR 3000 above it (1.23); neither comparison is meaningful, because the estimator is bounded and biased. What the control does show is that the bound itself is of order one in log units: 2.1 in $\log T_1$ at SNR 1000, i.e. a one-sigma factor of $e^{2.1} \approx 8$, and still 0.70 (a factor 2) at SNR 3000, while T_2 is unaffected. With small angles, B_1^+ and T_1 cannot

Table 4: Monte Carlo errors of $\log \hat{T}_1$ (1000 noise realisations, reference tissue, $B_1^+ = 1$). SD, mean (bias), RMSE and robust spread (IQR/1.349) are over voxels with a defined estimate. “Undef.”: fraction with no valid T_1 ; “bound”: fraction with a constraint or search limit active (for ML: $R'_2 = 0$ or the safeguard box). CRLB: bound on the SD for unbiased estimators using both scans with B_1^+ unknown.

Protocol	SNR(M_0)	Estimator	SD	mean	RMSE	robust	undef. %	bound %
15°/330°	300	CRLB	0.135					
		Joint ML	0.128	-0.007	0.128	0.134	0.0	19.5
		Model fit	0.939	+0.335	0.996	0.907	0.0	17.6
		Analytic inverse	0.959	+0.445	1.056	1.026	1.9	0.2
		Analytic, B_1^+ given	0.526	+0.119	0.539	0.359	1.9	–
15°/330°	1000	CRLB	0.041					
		Joint ML	0.041	+0.001	0.041	0.039	0.0	0.3
		Model fit	0.157	+0.016	0.158	0.114	0.0	0.0
		Analytic inverse	0.207	+0.036	0.210	0.127	0.0	0.0
		Analytic, B_1^+ given	0.108	+0.009	0.108	0.107	0.0	–
15°/330°	3000	CRLB	0.014					
		Joint ML	0.014	+0.000	0.014	0.013	0.0	0.0
		Model fit	0.038	+0.001	0.038	0.040	0.0	0.0
		Analytic inverse	0.043	+0.001	0.043	0.044	0.0	0.0
		Analytic, B_1^+ given	0.035	+0.000	0.035	0.039	0.0	–
15°/30°	1000	CRLB	2.106					
		Joint ML	1.364	+0.409	1.423	2.117	0.0	34.8
		Model fit	1.794	+0.094	1.796	2.876	0.0	66.2
15°/30°	3000	CRLB	0.702					
		Joint ML	0.861	+0.244	0.894	0.771	0.0	10.2
		Model fit	1.492	+0.267	1.515	2.256	0.0	34.7

be separated usefully at these SNRs by any unbiased estimator; this is the forward model and protocol (Corollary 2).

Sources of the estimator-specific loss. The analytic inverse (i) is *sequential*: R_2 and R'_2 are fixed from scan 1 before the flip angle and T_1 are estimated, so their errors propagate instead of being traded off jointly; (ii) uses *closed-form nonlinear transformations* (differences of extrapolated states, ratios, logarithms) whose noise amplification is not controlled and whose domain can be left by noise; (iii) computes T_1 from *scan 1* only, with B_1^+ held fixed; and (iv) discards phase (Eq. 18), which costs little here because $\Delta\omega$ and ϕ_0 decouple from the other parameters (Fig. 1).

Point (iii) combines two separate changes relative to the joint bound of 40.5, which the table in Section 4 separates: treating B_1^+ as known (40.5 to 28.0 with both scans) and dropping scan 2 from the T_1 step (28.0 to 57.1 with B_1^+ known). The matched reference for the analytic inverse with B_1^+ given is therefore 57.1, not 40.5 or 28.0 (Table 6). In addition, holding B_1^+ fixed turns a B_1^+ error ε into a T_1 bias of -2.02ε in this step (Table 3). Dropping scan 2 is partly an acquisition choice: scan 2 is acquired only at low resolution, so a full-resolution bound using both scans is not available to any estimator in the published design. At the low resolution both scans have about four times the per-voxel SNR, giving a B_1^+ bound of $2.91/4 \approx 0.73$ per low-resolution voxel before spatial smoothing.

Table 5: Monte Carlo SD (robust spread in parentheses) of $\log \hat{B}_1^+$, $\log \hat{T}_2$ and $\log \hat{M}_0$, and the mean of $\log \hat{B}_1^+$. Same runs as Table 4.

Protocol	SNR	Estimator	B_1^+ SD (rob.)	B_1^+ mean	T_2 SD (rob.)	M_0 SD (rob.)
15°/330°	300	CRLB	0.010		0.077	0.063
		Joint ML	0.010 (0.009)	-0.0017	0.075 (0.074)	0.063 (0.060)
		Model fit	0.060 (0.037)	+0.0011	0.476 (0.368)	0.649 (0.569)
		Analytic inverse	0.270 (0.461)	-0.1699	0.506 (0.370)	0.604 (0.571)
15°/330°	1000	CRLB	0.003		0.023	0.019
		Joint ML	0.003 (0.003)	-0.0001	0.023 (0.025)	0.019 (0.019)
		Model fit	0.017 (0.013)	-0.0025	0.115 (0.110)	0.108 (0.085)
		Analytic inverse	0.066 (0.015)	-0.0135	0.118 (0.112)	0.118 (0.076)
15°/330°	3000	CRLB	0.001		0.008	0.006
		Joint ML	0.001 (0.001)	+0.0000	0.008 (0.008)	0.006 (0.006)
		Model fit	0.004 (0.004)	-0.0003	0.036 (0.038)	0.028 (0.029)
		Analytic inverse	0.005 (0.004)	-0.0005	0.037 (0.037)	0.025 (0.026)
15°/30°	1000	CRLB	1.023		0.023	1.048
		Joint ML	0.663 (1.037)	-0.2099	0.022 (0.022)	0.679 (1.053)
		Model fit	0.862 (1.380)	-0.0678	0.115 (0.110)	0.937 (1.490)
15°/30°	3000	CRLB	0.341		0.008	0.349
		Joint ML	0.422 (0.373)	-0.1228	0.007 (0.007)	0.429 (0.381)
		Model fit	0.723 (1.097)	-0.1443	0.036 (0.038)	0.757 (1.139)

Table 6: Joint ML diagnostics, and the analytic inverse with B_1^+ given compared with matched bounds: both scans with B_1^+ known (28.0 per unit σ/M_0) and scan 1 only with B_1^+ known (57.1), the data its T_1 step actually uses.

Protocol	SNR	Joint ML			Analytic, B_1^+ given: SD / bound	
		$R'_2 = 0\%$	box %	cost > cost(truth)	both scans	scan 1 only
15°/330°	300	19.5	0.0	0	5.65	2.77
15°/330°	1000	0.3	0.0	0	3.86	1.89
15°/330°	3000	0.0	0.0	0	3.80	1.86
15°/30°	1000	0.1	34.7	0	—	—
15°/30°	3000	0.0	10.2	0	—	—

Table 7: Origin of each difficulty.

Difficulty	Origin	Remedy
Leading-order M_0 - B_1^+ - T_1 invariance (21) for small flip angles (severe ill-conditioning, not exact singularity)	forward model + protocol	a pulse whose effective angle does not scale with B_1^+ (e.g. near 360° , as published); otherwise very high SNR
Residual T_1 bound (40.5 per unit σ/M_0): confounding with T_2 , T_2^* , M_0 , B_1^+	forward model + protocol	SNR, averaging, spatial priors; protocol tuning ($\leq 20\%$)
Exact magnitude alias $\alpha_2 = 360^\circ \mp x$ at physiological B_1^+ (Section 6)	forward model: magnitudes lose the sign of $\sin \alpha$	FID-sign test, or complex data with a shared phase
Exact complex alias $\alpha_2 > 540^\circ$ ($B_1^+ \approx 2$)	forward model: 2π periodicity	range assumption $B_1^+ < 1.64$
Error well above the bound; large bias and undefined T_1 at moderate SNR	analytic inverse (sequential, closed-form)	joint (ML or learned) estimation
T_1 bound 57.1 instead of 28.0 with B_1^+ known; T_1 bias -2ε per B_1^+ error ε	design: scan-1-only T_1 step, low-resolution scan 2	joint estimation with a smooth- B_1^+ prior
Constrained estimates ($R_2' = 0$) at low SNR	forward model domain + small true R_2'	report; Bayesian or constrained bounds
In-vivo calibration $\beta = 1.24$ (Eq. 19 of [1])	model mismatch (not variance)	richer forward model (finite pulses, MT, ...)

6 Magnitude data and the flip-angle branch

The analytic inverse of [1] works with magnitudes, made real-valued by the sign convention of their Eq. 18. This section asks how much information magnitudes carry, how a direct least-squares fit to magnitudes compares with the analytic inverse, and why a magnitude fit needs an explicit branch test that complex maximum likelihood does not.

6.1 Information in magnitude data

With complex Gaussian noise of standard deviation σ per component, a magnitude $m = |S + n|$ is Rician with amplitude $A = |S|$. Its Fisher information about A is $g(A/\sigma)/\sigma^2$, where

$$g(a) = \mathbb{E} \left[x^2 \frac{I_1(xa)^2}{I_0(xa)^2} \right] - a^2, \quad x = m/\sigma, \quad (32)$$

with I_0, I_1 modified Bessel functions. g rises from 0 at $a = 0$ to 1 as $a \rightarrow \infty$ (e.g. $g(0.5) = 0.20$, $g(2) = 0.85$, $g(5) = 0.98$). The information about θ is

$$\mathcal{I}_{\text{mag}} = \frac{1}{\sigma^2} \sum_j g(A_j/\sigma) \nabla_{\theta} A_j \nabla_{\theta} A_j^{\top}, \quad \nabla_{\theta} A_j = \text{Re}(\overline{S_j} \nabla_{\theta} S_j) / A_j, \quad (33)$$

over the five magnitude parameters ($\Delta\omega$ and ϕ_0 do not affect magnitudes). Unlike the complex case it is not simply proportional to $1/\sigma^2$, so it is evaluated at each SNR.

For the published protocol and the reference tissue the weakest measured amplitude is $0.0074 M_0$ (scan 2, pathway +1), so even at $\text{SNR}(M_0) = 300$ every measurement has $A/\sigma \geq 2.2$ and $g \geq 0.85$. The magnitude bounds equal the complex bounds to within 0.5% for T_1 , B_1^+ , T_2 and M_0 at SNR 300, 1000 and 3000 (Table 8). Discarding the phase costs essentially no *local* information, because the phase mainly carries $\Delta\omega$ and ϕ_0 , which decouple from the other parameters (Fig. 1). What the phase does carry is a single *global* bit, discussed next.

6.2 Why magnitude fits need the branch test and complex ML does not

1. Magnitudes lose the sign of $\sin \alpha$. By Theorem 1 the magnitudes of scan i determine u_i and $M_0|v_i|$; by (17) the latter carries no information beyond u_i and T_1 . The data therefore fix $\cos \alpha_i$ for any trial E_1 , but not the sign of $\sin \alpha_i$. Concretely, $|a_k(\alpha)|$ is even and 2π -periodic in α , hence symmetric about every multiple of 180° .

2. This produces an exact alias inside the physiological range. For scan 2, $\alpha_2 = 360^\circ - x$ (the truth for $B_1^+ < 1.09$) and $\alpha_2 = 360^\circ + x'$ give magnitudes of the same form. A different B_1^+ also changes α_1 , but Theorem 1 shows that a change of α_1 can be absorbed by T_1 and M_0 through $(u_1, M_0 v_1)$. The result is an exact alias (Table 1): for the reference tissue, $(B_1^+, T_1, M_0) = (1, 1500 \text{ ms}, 1)$ and $(1.199, 1038 \text{ ms}, 0.832)$ give the same 18 magnitudes to machine precision. The alias is close to the small-angle scaling $T_1 \propto (B_1^+)^{-2}$, $M_0 \propto (B_1^+)^{-1}$ ($1500/1.199^2 = 1043$), but exact.

3. It is a symmetry of the magnitude likelihood, not a noise effect. Because the alias exists for every parameter value (all rows of Table 1), the map $\theta \mapsto |s(\theta)|$ takes the same values on two separate sheets of parameter space. Any likelihood that depends on the data only through magnitudes, Gaussian or Rician, then has two equal global maxima for *every* noise realisation. We

checked this directly: fitting noisy magnitudes on each side of 360° gives final costs that agree to a median relative difference of 10^{-12} at SNR 300, 1000 and 3000. No SNR resolves it, and the local CRLB, which is identical for both sheets, cannot detect it. An unconstrained magnitude fit simply returns the maximum nearest its starting point. In our Monte Carlo it returned the wrong branch in 47, 3 and 0 of 1000 voxels at SNR 300, 1000 and 3000; these counts describe the starting points, not the data.

4. Complex data carry the missing bit. The complex model keeps the sign: $a_k(-\alpha) = -a_k(\alpha)$ for all k (Proposition 3), so moving α_2 from $360^\circ - x$ to $360^\circ + x'$ flips the sign of every scan-2 signal relative to scan 1. A single phase ϕ_0 (and a single $\Delta\omega$) is shared by both scans, so this relative sign cannot be absorbed. The best complex fit restricted to $360^\circ < \alpha_2 < 540^\circ$ moves to the edge $\alpha_2 = 360^\circ$, where scan 2 vanishes, and leaves the entire scan-2 signal as residual ($0.144 M_0$). Its cost exceeds the true-branch cost by $1.9 \times 10^3 \sigma^2$ at SNR 300 ($2.1 \times 10^4 \sigma^2$ at SNR 1000), a likelihood ratio of about e^{-935} or smaller: the complex likelihood has no competing maximum on that branch, and complex ML selects the correct branch automatically.

5. The FID sign test restores exactly that bit. The test of [1],

$$\text{sign Re} \sum_j S_{1j0} \overline{S_{2j0}} = \text{sign}(\sin \alpha_1 \sin \alpha_2), \quad (34)$$

uses the relative phase of the two FID images, and only its sign. It is the one bit that magnitude processing discards and complex ML uses implicitly. We implement it in the magnitude fit as a constraint: B_1^+ is confined to the interval between consecutive zeros of $\sin(B_1^+ \alpha_i^{\text{nom}})$ on which $\sin \alpha_1 \sin \alpha_2$ has the measured sign; a start on the wrong side is reflected across the nearest zero (where $|a_k|$ is symmetric), and every step is projected back into the interval.

6. What the test cannot do. Branches with the same sign pattern (the complex aliases of Table 1, $\alpha_2 > 540^\circ$) are invisible to both the FID test and complex ML; they are excluded only by the range assumption $B_1^+ < 1.64$. Near $\alpha_2^{\text{eff}} = 360^\circ$ the scan-2 FID vanishes, the sign is unreliable, and the two branches merge, so a wrong decision there costs little. Finally, both the test and complex ML require that the relative phase of the two scans is preserved: the same receiver phase and coil combination, and no motion or B_0 drift between the sequential acquisitions. If each scan had its own unknown phase, complex ML would need a phase per scan, the relative sign would be lost, and the magnitude alias would reappear for complex data too.

6.3 Comparison on identical data

We compare, on the same noise realisations as Table 4: the analytic inverse; a Gaussian least-squares fit to the magnitudes, $\min_\theta \sum_j (|y_j| - |s_j(\theta)|)^2$, with the same projected Levenberg–Marquardt iteration and four starts as the ML fit, without and with the FID-sign constraint; and complex ML. Magnitude least squares is *not* maximum likelihood for Rician data, because it ignores the noise floor $\mathbb{E}|y| - A \approx \sigma^2/(2A)$; for the weakest amplitude this is 10% at SNR 300, 1% at SNR 1000 and 0.1% at SNR 3000.

Reading Table 8.

- **With the branch test, magnitude least squares performs like complex ML.** Its SD/CRLB for T_1 and B_1^+ is within the Monte Carlo error of complex ML at every SNR, and

Table 8: Magnitude versus complex estimation on identical data (paper protocol, reference tissue, 1000 realisations). SD and mean of the log-estimates; bounds for complex data and for magnitude (Rician) data. Last column: voxels on the wrong side of $\alpha_2 = 360^\circ$ (“wrong”).

SNR		T_1 SD	T_1 mean	B_1^+ SD	T_2 SD	wrong
300	CRLB complex/Rician	0.1350/0.1356		0.0097/0.0098	0.0769/0.0771	
	Analytic inverse [1]	0.9587	+0.445	0.2695	0.5060	–
	Magnitude LS	0.1909	-0.030	0.0692	0.0743	47
	Magnitude LS + FID sign	0.1251	-0.005	0.0094	0.0742	0
	Complex ML	0.1281	-0.007	0.0097	0.0749	–
1000	CRLB complex/Rician	0.0405/0.0405		0.0029/0.0029	0.0231/0.0231	
	Analytic inverse [1]	0.2072	+0.036	0.0657	0.1180	–
	Magnitude LS	0.0457	+0.000	0.0104	0.0231	3
	Magnitude LS + FID sign	0.0409	+0.001	0.0029	0.0231	0
	Complex ML	0.0411	+0.001	0.0029	0.0231	–
3000	CRLB complex/Rician	0.0135/0.0135		0.0010/0.0010	0.0077/0.0077	
	Analytic inverse [1]	0.0425	+0.001	0.0046	0.0373	–
	Magnitude LS	0.0136	+0.000	0.0010	0.0076	0
	Magnitude LS + FID sign	0.0136	+0.000	0.0010	0.0076	0
	Complex ML	0.0136	+0.000	0.0010	0.0076	–

no voxel ends on the wrong branch. At SNR 300 the Rician floor adds a small bias (-0.005 in $\log T_1$), and, as for complex ML, the constraint $R_2' \geq 0$ is active in about a fifth of the voxels.

- **Without it, wrong-branch voxels dominate the B_1^+ error.** Their robust spread is at the bound, but the standard deviation is up to seven times larger, and the wrong branch also biases T_1 by a factor of about $(B_1^+)^{-2}$.
- **The analytic inverse is the outlier.** It uses the same magnitudes and the same FID sign, yet its errors are 3–7 times larger for T_1 and up to 28 times for B_1^+ (Table 4). Since magnitudes carry essentially all the local information and a joint magnitude fit extracts it, the loss is entirely algorithmic: the sequential, closed-form structure of the inversion, not the decision to use magnitudes.

7 Exact Jacobians by implicit differentiation

Both least-squares estimators (complex ML, Section 5.1, and the magnitude fit, Section 6) spend nearly all their time forming Jacobians. The forward-difference version evaluates the model once per parameter at every iteration. This section derives the exact Jacobian from the structure of the steady state and shows that it is an order of magnitude faster, with identical estimates.

7.1 Which differentiation mode

Per voxel the Jacobian is 36×7 for complex data (18 complex values, real and imaginary parts) and 18×5 for magnitudes. Reverse-mode differentiation (back-propagation) produces one *row* per backward pass, the gradient of one scalar output; even with all voxels batched in one pass (voxels are independent), it needs 36 or 18 passes. Forward mode produces one *column* per pass, so 7 or 5 passes, each about as costly as a model evaluation. Because there are fewer parameters than outputs, forward mode is the natural choice, and finite differences are a (less accurate) forward mode. Exploiting the structure of the model does better than either.

7.2 Differentiating the steady state

With RF phase 0 and intravoxel phase ψ , the post-pulse steady state of one isochromat solves the 3×3 linear system (6),

$$A \mathbf{M} = \mathbf{b}, \quad A = I - RQ, \quad Q = R_z(\psi) \text{diag}(E_2, E_2, E_1), \quad \mathbf{b} = (1 - E_1) R \hat{\mathbf{z}}, \quad (35)$$

with $R = R_x(\alpha)$. Off-resonance enters only as a phase factor (Proposition 1), so the steady state is computed at $\Delta\omega = 0$. Differentiating $A \mathbf{M} = \mathbf{b}$ with respect to any $x \in \{\alpha, E_1, E_2\}$ gives

$$\frac{\partial \mathbf{M}}{\partial x} = A^{-1} \left(\frac{\partial \mathbf{b}}{\partial x} - \frac{\partial A}{\partial x} \mathbf{M} \right), \quad (36)$$

which needs no new matrix inversion: the same A^{-1} serves the solution and all three derivatives. The ingredients are closed form. With $R' = \partial R_x / \partial \alpha$ and $\Pi_z = \text{diag}(0, 0, 1)$, $\Pi_{xy} = \text{diag}(1, 1, 0)$ (note $R_z \Pi_z = \Pi_z$):

$$\begin{aligned} \frac{\partial A}{\partial \alpha} &= -R'Q, & \frac{\partial \mathbf{b}}{\partial \alpha} &= (1 - E_1) R' \hat{\mathbf{z}}, \\ \frac{\partial A}{\partial E_1} &= -R \Pi_z, & \frac{\partial \mathbf{b}}{\partial E_1} &= -R \hat{\mathbf{z}}, \\ \frac{\partial A}{\partial E_2} &= -R R_z(\psi) \Pi_{xy}, & \frac{\partial \mathbf{b}}{\partial E_2} &= \mathbf{0}. \end{aligned} \quad (37)$$

The three right-hand sides of (36) are therefore

$$(1 - E_1) R' \hat{\mathbf{z}} + R' Q \mathbf{M}, \quad -R \hat{\mathbf{z}} + R \hat{\mathbf{z}} M_z, \quad R R_z(\psi) \Pi_{xy} \mathbf{M}. \quad (38)$$

Inverting 3×3 matrices. There are $N \times n_{\text{iso}}$ such systems per scan (e.g. $65\,536 \times 128$ for one slice). Batched general-purpose solvers carry a large per-matrix overhead at this size, so we form A^{-1} explicitly by the adjugate formula, $A^{-1} = \text{adj}(A) / \det A$, which is nine cofactors and one determinant computed elementwise over the whole batch. A is well conditioned for physical parameters ($\det A \neq 0$ whenever $E_1, E_2 < 1$).

Pathways and magnitudes. The pathway amplitudes are discrete Fourier coefficients, over the isochromats, of the transverse magnetisation $m_{\perp}(\psi_n)$. The transform is linear, so $\partial a_k / \partial x$ is the same FFT applied to $\partial m_{\perp} / \partial x$. For magnitude data,

$$\frac{\partial |a_k|}{\partial x} = \frac{\text{Re}(\overline{a_k} \partial a_k / \partial x)}{|a_k|}.$$

7.3 Chain rule to the fitted parameters

With $\Phi = \phi_0 + \Delta\omega \tau$, $\tau = t + kT_R$, $D = \exp(-t/T_2 - R'_2|\tau|)$ and $R'_2 = 1/T_2^* - 1/T_2$, the signal is $S = M_0 a_k D e^{i\Phi}$, and $\partial \alpha / \partial \log B_1^+ = \alpha$, $\partial E_1 / \partial \log T_1 = E_1 T_R / T_1$, $\partial E_2 / \partial \log T_2 = E_2 T_R / T_2$. Then

$$\begin{aligned} \frac{\partial S}{\partial \log M_0} &= S, & \frac{\partial S}{\partial \log B_1^+} &= M_0 D e^{i\Phi} \alpha \frac{\partial a_k}{\partial \alpha}, \\ \frac{\partial S}{\partial \log T_1} &= M_0 D e^{i\Phi} \frac{E_1 T_R}{T_1} \frac{\partial a_k}{\partial E_1}, & \frac{\partial S}{\partial \log T_2} &= M_0 D e^{i\Phi} \frac{E_2 T_R}{T_2} \frac{\partial a_k}{\partial E_2} + S \frac{t - |\tau|}{T_2}, \\ \frac{\partial S}{\partial \log T_2^*} &= S \frac{|\tau|}{T_2^*}, & \frac{\partial S}{\partial \Delta\omega} &= i\tau S, & \frac{\partial S}{\partial \phi_0} &= iS. \end{aligned} \quad (39)$$

The second term of $\partial S/\partial \log T_2$ combines the echo decay e^{-t/T_2} and the coupling through R'_2 ($\partial R'_2/\partial \log T_2 = 1/T_2$). For the magnitude model the same expressions hold with $|a_k|$ in place of a_k and without the phase factor and the last two derivatives.

7.4 Levenberg–Marquardt with exact Jacobians

A forward-difference iteration costs one trial evaluation plus one evaluation per parameter (8 for complex data, 6 for magnitudes). With exact Jacobians each trial evaluation returns the residual *and* its Jacobian. If the step is accepted, that Jacobian is the one needed for the next iteration; if it is rejected, the current Jacobian is kept and only λ changes. Each iteration therefore costs one model-plus-Jacobian evaluation, i.e. one explicit inverse and four right-hand sides per isochromat. Exact derivatives also make single precision usable: forward differences in float32 would need steps of order 10^{-3} and give inaccurate Jacobians, whereas the implicit Jacobian is accurate to about 10^{-6} in float32. A small Jacobian error affects only the convergence path, not the solution, which is determined by the residual.

7.5 Verification and benchmark

On random parameters the implicit Jacobians agree with automatic differentiation of the reference model to 6×10^{-15} (magnitude) and 3×10^{-16} (complex) in double precision, and to 10^{-6} in single precision; both are unit tests. Table 9 compares wall times on 4096 voxels with random tissue (T_1 600–3000 ms, T_2 40–120 ms, B_1^+ 0.8–1.1), the paper protocol, $\text{SNR}(M_0) = 1000$, one start and 15 iterations (enough: 15 and 40 iterations differ by less than 10^{-9} in $\log T_1$), on a laptop CPU with 8 threads and no GPU.

Table 9: Fitting time per voxel and for a 256×256 slice, and agreement with the forward-difference float64 reference (largest $|\Delta \log \hat{T}_1|$ over the 4096 voxels). The noise-induced spread of $\log \hat{T}_1$ is 0.164 in every magnitude variant and 0.163 in every complex variant, so all differences are negligible.

Fit	Jacobian, precision, isochromats	ms/voxel	256 ² slice	speed-up	max $ \Delta \log \hat{T}_1 $
Magnitude	finite differences, f64, 256	15.0	16.4 min	1.0	reference
	implicit, f64, 256	1.41	1.5 min	10.7	1.6×10^{-5}
	implicit, f64, 128	0.73	0.8 min	20.6	1.6×10^{-5}
	implicit, f32, 256	0.74	0.8 min	20.2	1.2×10^{-3}
	implicit, f32, 128	0.44	0.5 min	33.7	1.0×10^{-3}
Complex ML	finite differences, f64, 256	19.2	21.0 min	1.0	reference
	implicit, f64, 256	1.55	1.7 min	12.4	1.4×10^{-5}
	implicit, f64, 128	0.83	0.9 min	23.2	1.4×10^{-5}
	implicit, f32, 256	0.90	1.0 min	21.3	1.2×10^{-3}
	implicit, f32, 128	0.56	0.6 min	34.4	1.2×10^{-3}

The exact Jacobian alone gives a factor of 10–12, more than the reduction in model evaluations (from 6–8 to one per iteration) would suggest, because the explicit adjugate inverse also avoids the per-matrix overhead of the general solver. Halving the number of isochromats and moving to single precision each roughly halve the time again; neither changes the estimates by more than 10^{-3} in $\log \hat{T}_1$, less than 1% of the noise-induced spread. The remaining differences against the reference, of order 10^{-5} in double precision, come from the inexact forward-difference Jacobian, which converges slightly differently in a few voxels. A 256×256 slice now fits in about half a minute and a whole brain (about 160 slices) in about 80 minutes on this CPU with one start, or about half that with a

brain mask. Multiple starts multiply these times; with the branch fixed by the FID sign, one start plus a check of the final cost is likely sufficient, but this has not yet been tested at low SNR.

8 Implications

1. The published protocol is sound from an information standpoint: the near-360° pulse lifts the leading-order invariance that makes B_1^+/T_1 separation extremely noise-sensitive with small angles.
2. No *unbiased* voxel-wise estimator can beat Table 2. The realistic voxel-wise gain of an unbiased estimator over the analytic inverse is the gap that the ML fit closes (Table 4). Biased estimators, including learned ones trained with a prior over tissue parameters, can have lower variance or MSE than the bound; their benefit then comes from the prior, and their bias must be characterised (for example with a Bayesian bound) rather than ignored.
3. Larger gains require prior information across voxels. B_1^+ is spatially smooth, so treating it as nearly known removes its factor 1.45; the remaining confounding with M_0 and the transverse relaxation times can only be reduced by image priors.
4. When B_1^+ is supplied to a joint estimator that also uses the high-flip data, a B_1^+ error is amplified about tenfold into T_1 (Table 3); B_1^+ should be estimated jointly or the high-flip data excluded from the T_1 step.

Caveats. The CRLB is local and assumes an exact model, Gaussian noise, an interior true parameter and (locally) unbiased estimators; constrained or biased estimates fall outside it. The model uses instantaneous RF pulses (no off-resonance nutation, Eq. 8 of [1]), a Lorentzian R_2' line shape, no diffusion and no magnetisation transfer. The time from the pulse to the first echo is assumed. Except where stated, both scans are treated at full resolution.

A Reproducibility

Forward model	<code>src/mpme/epg.py, signal.py, sequence.py</code>
Analytic inverse of [1]	<code>src/mpme/paper.py</code>
Model-fit baseline	<code>src/mpme/analytic.py</code>
Joint ML fit, magnitude fit, branch test	<code>src/mpme/mle.py</code>
Rician information	<code>src/mpme/crlb.py</code>
Implicit Jacobians	<code>src/mpme/fastjac.py</code>
Table 9	<code>python scripts/benchmark_fit.py</code>
Fisher information, CRLB	<code>src/mpme/crlb.py</code>
Tables 2–3, figures	<code>python scripts/crlb_analysis.py</code>
Tables 4–6	<code>python scripts/estimator_efficiency.py</code>
Table 8	<code>python scripts/magnitude_comparison.py</code>

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